

DATA INTERPRETATION AND ANALYSIS

Full Question Bank
Questions, Solutions, and Explanations

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SECTION: 01 Basic

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--- README.md ---

100 Data Interpretation Practice Questions (Basic Level) with Solutions

This file contains 100 basic-level practice questions for Data Interpretation, designed to develop your ability to analyze and interpret data for fresher job aptitude tests, especially for AI/ML roles. Each question includes a dataset (tables, bar graphs, pie charts, line graphs, or caselets) and a detailed solution with step-by-step calculations. Topics cover basic arithmetic operations, percentages, ratios, averages, and simple comparisons.

Table-Based Questions (1?20)

Table 1: Monthly Sales (in units) of Five Products in a Store (Jan?Apr 2023)

Product	Jan	Feb	Mar	Apr
A	200	250	300	350
B	150	180	200	220
C	100	120	140	160
D	300	280	260	240
E	250	270	290	310

1. Question: What is the total sales of Product A across all months?

Solution:

Total sales = $200 + 250 + 300 + 350 = 1100$ units.

Answer: 1100 units

2. Question: Which product had the highest sales in February?

Solution:

Feb sales: A (250), B (180), C (120), D (280), E (270). Highest is D (280).

Answer: Product D

3. Question: What is the average monthly sales of Product B?

Solution:

Total sales = $150 + 180 + 200 + 220 = 750$. Average = $750 \div 4 = 187.5$ units.

Answer: 187.5 units

4. Question: By how much did Product C's sales increase from Jan to Apr?

Solution:

Increase = Apr (160) - Jan (100) = 60 units.

Answer: 60 units

5. Question: What is the total sales of all products in March?

Solution:

Mar sales: A (300) + B (200) + C (140) + D (260) + E (290) = 1190 units.

Answer: 1190 units

6. Question: Which product had the lowest sales in April?

Solution:

Apr sales: A (350), B (220), C (160), D (240), E (310). Lowest is C (160).

Answer: Product C

7. Question: What is the percentage increase in Product A's sales from Jan to Feb?

Solution:

Increase = 250 - 200 = 50. % Increase = $(50 \div 200) \times 100 = 25\%$.

Answer: 25%

8. Question: What is the ratio of Product B's sales in Feb to Mar?

Solution:

Feb (180) : Mar (200) = 180 : 200 = 9 : 10.

Answer: 9 : 10

9. Question: What is the total sales of Products D and E in Jan?

Solution:

Jan sales: D (300) + E (250) = 550 units.

Answer: 550 units

10. Question: Which month had the highest total sales?

Solution:

Total sales: Jan (1000), Feb (1100), Mar (1190), Apr (1280). Highest is Apr.

Answer: April

Table 2: Number of Students in Five Classes (A-E) in Two Years

Class	2022	2023
A	40	50
B	30	35
C	50	45
D	60	70
E	20	25

11. Question: What is the total number of students in Class D across both years?

Solution:

$$\text{Total} = 60 + 70 = 130 \text{ students.}$$

Answer: 130 students

12. Question: Which class had the highest number of students in 2023?

Solution:

2023: A (50), B (35), C (45), D (70), E (25). Highest is D (70).

Answer: Class D

13. Question: What is the average number of students in Class A over the two years?

Solution:

$$\text{Total} = 40 + 50 = 90. \text{ Average} = 90 \div 2 = 45 \text{ students.}$$

Answer: 45 students

14. Question: By how much did Class B's student count increase from 2022 to 2023?

Solution:

$$\text{Increase} = 35 - 30 = 5 \text{ students.}$$

Answer: 5 students

15. Question: What is the total number of students in 2022 across all classes?

Solution:

$$\text{Total} = 40 + 30 + 50 + 60 + 20 = 200 \text{ students.}$$

Answer: 200 students

16. Question: Which class had the lowest number of students in 2022?

Solution:

2022: A (40), B (30), C (50), D (60), E (20). Lowest is E (20).

Answer: Class E

17. Question: What is the percentage increase in Class A's students from 2022 to 2023?

Solution:

$$\text{Increase} = 50 - 40 = 10. \text{ \% Increase} = (10 \div 40) \times 100 = 25\%.$$

Answer: 25%

18. Question: What is the ratio of students in Class C in 2022 to 2023?

Solution:

$$2022 (50) : 2023 (45) = 50 : 45 = 10 : 9.$$

Answer: 10 : 9

19. Question: What is the total number of students in Classes A and B in 2023?

Solution:

$$2023: A (50) + B (35) = 85 \text{ students.}$$

Answer: 85 students

20. Question: Which year had more total students?

Solution:

2022: 200 students. 2023: $50 + 35 + 45 + 70 + 25 = 225$ students. 2023 is higher.

Answer: 2023

Bar Graph-Based Questions (21-40)

Bar Graph 1: Annual Revenue (in \$1000) of Four Companies (2019-2021)

- 2019: P (50), Q (40), R (60), S (30)

- 2020: P (55), Q (45), R (65), S (35)

- 2021: P (60), Q (50), R (70), S (40)

21. Question: What is Company P's total revenue over the three years?

Solution:

Total = $50 + 55 + 60 = 165$ (\$1000).

Answer: \$165,000

22. Question: Which company had the highest revenue in 2020?

Solution:

2020: P (55), Q (45), R (65), S (35). Highest is R (65).

Answer: Company R

23. Question: What is the average revenue of Company Q over the three years?

Solution:

Total = $40 + 45 + 50 = 135$. Average = $135 \div 3 = 45$ (\$1000).

Answer: \$45,000

24. Question: By how much did Company S's revenue increase from 2019 to 2021?

Solution:

Increase = $40 - 30 = 10$ (\$1000).

Answer: \$10,000

25. Question: What is the total revenue of all companies in 2019?

Solution:

2019: $50 + 40 + 60 + 30 = 180$ (\$1000).

Answer: \$180,000

26. Question: Which company had the lowest revenue in 2021?

Solution:

2021: P (60), Q (50), R (70), S (40). Lowest is S (40).

Answer: Company S

27. Question: What is the percentage increase in Company P's revenue from 2019 to 2020?

Solution:

Increase = $55 - 50 = 5$. % Increase = $(5 \div 50) \times 100 = 10\%$.

Answer: 10%

28. Question: What is the ratio of Company Q's revenue in 2020 to 2021?

Solution:

$$2020 (45) : 2021 (50) = 45 : 50 = 9 : 10.$$

Answer: 9 : 10

29. Question: What is the total revenue of Companies R and S in 2020?

Solution:

$$2020: R (65) + S (35) = 100 (\$1000).$$

Answer: \$100,000

30. Question: Which year had the highest total revenue?

Solution:

2019: 180. 2020: $55 + 45 + 65 + 35 = 200$. 2021: $60 + 50 + 70 + 40 = 220$ (\$1000). Highest is 2021.

Answer: 2021

Bar Graph 2: Number of Books Sold by Four Genres (2022-2023)

- 2022: Fiction (200), Non-Fiction (150), Mystery (100), Sci-Fi (50)

- 2023: Fiction (220), Non-Fiction (160), Mystery (120), Sci-Fi (60)

31. Question: What is the total number of Fiction books sold over the two years?

Solution:

$$\text{Total} = 200 + 220 = 420 \text{ books.}$$

Answer: 420 books

32. Question: Which genre sold the most books in 2023?

Solution:

2023: Fiction (220), Non-Fiction (160), Mystery (120), Sci-Fi (60). Highest is Fiction.

Answer: Fiction

33. Question: What is the average number of Mystery books sold over the two years?

Solution:

$$\text{Total} = 100 + 120 = 220. \text{ Average} = 220 \div 2 = 110 \text{ books.}$$

Answer: 110 books

34. Question: By how many books did Sci-Fi sales increase from 2022 to 2023?

Solution:

$$\text{Increase} = 60 - 50 = 10 \text{ books.}$$

Answer: 10 books

35. Question: What is the total number of books sold in 2022?

Solution:

$$2022: 200 + 150 + 100 + 50 = 500 \text{ books.}$$

Answer: 500 books

36. Question: Which genre sold the fewest books in 2022?

Solution:

2022: Fiction (200), Non-Fiction (150), Mystery (100), Sci-Fi (50). Lowest is Sci-Fi.

Answer: Sci-Fi

37. Question: What is the percentage increase in Fiction sales from 2022 to 2023?

Solution:

Increase = $220 - 200 = 20$. % Increase = $(20 \div 200) \times 100 = 10\%$.

Answer: 10%

38. Question: What is the ratio of Non-Fiction sales in 2022 to 2023?

Solution:

2022 (150) : 2023 (160) = $150 : 160 = 15 : 16$.

Answer: 15 : 16

39. Question: What is the total number of Mystery and Sci-Fi books sold in 2023?

Solution:

2023: Mystery (120) + Sci-Fi (60) = 180 books.

Answer: 180 books

40. Question: Which year had more total book sales?

Solution:

2022: 500. 2023: $220 + 160 + 120 + 60 = 560$ books. 2023 is higher.

Answer: 2023

Pie Chart-Based Questions (41-60)

Pie Chart 1: Budget Allocation of a Company (Total: \$100,000)

- Salaries: 40%

- Rent: 20%

- Supplies: 15%

- Marketing: 15%

- Miscellaneous: 10%

41. Question: How much is allocated to Salaries?

Solution:

Salaries = 40% of 100,000 = $(40 \div 100) \times 100,000 = \$40,000$.

Answer: \$40,000

42. Question: What is the amount allocated to Rent?

Solution:

Rent = 20% of 100,000 = $(20 \div 100) \times 100,000 = \$20,000$.

Answer: \$20,000

43. Question: What is the total allocation for Supplies and Marketing?

Solution:

Supplies (15%) + Marketing (15%) = 30%. 30% of 100,000 = \$30,000.

Answer: \$30,000

44. Question: How much more is allocated to Salaries than to Miscellaneous?

Solution:

Salaries (\$40,000) - Miscellaneous (10% = \$10,000) = 40,000 - 10,000 = \$30,000.

Answer: \$30,000

45. Question: What percentage of the budget is allocated to Rent and Supplies combined?

Solution:

Rent (20%) + Supplies (15%) = 35%.

Answer: 35%

46. Question: Which category has the smallest allocation?

Solution:

Miscellaneous (10%) is the smallest.

Answer: Miscellaneous

47. Question: What is the ratio of Salaries to Marketing allocation?

Solution:

Salaries (40%) : Marketing (15%) = 40 : 15 = 8 : 3.

Answer: 8 : 3

48. Question: How much is allocated to Miscellaneous?

Solution:

Miscellaneous = 10% of 100,000 = \$10,000.

Answer: \$10,000

49. Question: What is the difference between Rent and Marketing allocations?

Solution:

Rent (20% = \$20,000) - Marketing (15% = \$15,000) = \$5,000.

Answer: \$5,000

50. Question: What is the total percentage allocated to Salaries and Rent?

Solution:

Salaries (40%) + Rent (20%) = 60%.

Answer: 60%

Pie Chart 2: Distribution of 500 Students by Department

- CS: 30%

- EE: 25%

- ME: 20%

- CE: 15%

- CH: 10%

51. Question: How many students are in the CS department?

Solution:

$$CS = 30\% \text{ of } 500 = (30 \div 100) \times 500 = 150 \text{ students.}$$

Answer: 150 students

52. Question: How many students are in the EE department?

Solution:

$$EE = 25\% \text{ of } 500 = (25 \div 100) \times 500 = 125 \text{ students.}$$

Answer: 125 students

53. Question: What is the total number of students in ME and CE?

Solution:

$$ME (20\%) + CE (15\%) = 35\%. 35\% \text{ of } 500 = (35 \div 100) \times 500 = 175 \text{ students.}$$

Answer: 175 students

54. Question: How many more students are in CS than in CH?

Solution:

$$CS (30\% = 150) - CH (10\% = 50) = 150 - 50 = 100 \text{ students.}$$

Answer: 100 students

55. Question: What percentage of students are in EE and ME combined?

Solution:

$$EE (25\%) + ME (20\%) = 45\%.$$

Answer: 45%

56. Question: Which department has the fewest students?

Solution:

CH (10%) is the smallest.

Answer: CH

57. Question: What is the ratio of CS to CE students?

Solution:

$$CS (30\%) : CE (15\%) = 30 : 15 = 2 : 1.$$

Answer: 2 : 1

58. Question: How many students are in the CH department?

Solution:

$$CH = 10\% \text{ of } 500 = (10 \div 100) \times 500 = 50 \text{ students.}$$

Answer: 50 students

59. Question: What is the difference between EE and ME student counts?

Solution:

$$EE (25\% = 125) - ME (20\% = 100) = 125 - 100 = 25 \text{ students.}$$

Answer: 25 students

60. Question: What is the total percentage of CS and EE students?

Solution:

$$\text{CS (30\%)} + \text{EE (25\%)} = 55\%.$$

Answer: 55%

Line Graph-Based Questions (61-80)

Line Graph 1: Monthly Temperature ($^{\circ}\text{C}$) in a City (Jan-Jun)

- Jan: 10

- Feb: 12

- Mar: 15

- Apr: 18

- May: 22

- Jun: 25

61. Question: What is the temperature in March?

Solution:

$$\text{Mar} = 15^{\circ}\text{C}.$$

Answer: 15°C

62. Question: What is the highest temperature recorded?

Solution:

Highest is Jun (25°C).

Answer: 25°C

63. Question: What is the average temperature over the six months?

Solution:

$$\text{Total} = 10 + 12 + 15 + 18 + 22 + 25 = 102. \text{ Average} = 102 \div 6 = 17^{\circ}\text{C}.$$

Answer: 17°C

64. Question: By how much did the temperature increase from Feb to Mar?

Solution:

$$\text{Increase} = 15 - 12 = 3^{\circ}\text{C}.$$

Answer: 3°C

65. Question: What is the total temperature for Jan and Feb?

Solution:

$$\text{Jan (10)} + \text{Feb (12)} = 22^{\circ}\text{C}.$$

Answer: 22°C

66. Question: Which month had the lowest temperature?

Solution:

Lowest is Jan (10°C).

Answer: January

67. Question: What is the percentage increase in temperature from Jan to Feb?

Solution:

$$\text{Increase} = 12 - 10 = 2. \quad \% \text{ Increase} = (2 \div 10) \times 100 = 20\%.$$

Answer: 20%

68. Question: What is the ratio of temperature in Mar to Apr?

Solution:

$$\text{Mar (15)} : \text{Apr (18)} = 15 : 18 = 5 : 6.$$

Answer: 5 : 6

69. Question: What is the temperature difference between May and Jun?

Solution:

$$\text{Jun (25)} - \text{May (22)} = 3^{\circ}\text{C}.$$

Answer: 3°C

70. Question: Which month had a temperature of 18°C?

Solution:

$$\text{Apr} = 18^{\circ}\text{C}.$$

Answer: April

Line Graph 2: Monthly Website Visitors (in thousands, Jan-Jun)

- Jan: 20

- Feb: 25

- Mar: 30

- Apr: 35

- May: 40

- Jun: 45

71. Question: How many visitors were there in April?

Solution:

$$\text{Apr} = 35 \text{ (thousand)} = 35,000 \text{ visitors}.$$

Answer: 35,000 visitors

72. Question: What is the highest number of visitors recorded?

Solution:

$$\text{Highest is Jun (45,000)}.$$

Answer: 45,000 visitors

73. Question: What is the average number of visitors over the six months?

Solution:

$$\text{Total} = 20 + 25 + 30 + 35 + 40 + 45 = 195. \quad \text{Average} = 195 \div 6 = 32.5 \text{ (thousand)} = 32,500.$$

Answer: 32,500 visitors

74. Question: By how many visitors did the count increase from Mar to Apr?

Solution:

Increase = $35 - 30 = 5$ (thousand) = 5,000 visitors.

Answer: 5,000 visitors

75. Question: What is the total number of visitors in Jan and Feb?

Solution:

Jan (20) + Feb (25) = 45 (thousand) = 45,000 visitors.

Answer: 45,000 visitors

76. Question: Which month had the fewest visitors?

Solution:

Lowest is Jan (20,000).

Answer: January

77. Question: What is the percentage increase in visitors from Feb to Mar?

Solution:

Increase = $30 - 25 = 5$. % Increase = $(5 \div 25) \times 100 = 20\%$.

Answer: 20%

78. Question: What is the ratio of visitors in May to Jun?

Solution:

May (40) : Jun (45) = $40 : 45 = 8 : 9$.

Answer: 8 : 9

79. Question: What is the difference in visitors between Apr and May?

Solution:

May (40) - Apr (35) = 5 (thousand) = 5,000 visitors.

Answer: 5,000 visitors

80. Question: Which month had 30,000 visitors?

Solution:

Mar = 30 (thousand) = 30,000.

Answer: March

Caselet-Based Questions (81?100)

Caselet 1: A Shop Sells Three Types of Fruits

- Apples: 100 kg sold at \$2/kg
- Bananas: 150 kg sold at \$1/kg
- Oranges: 200 kg sold at \$1.5/kg

81. Question: What is the total revenue from Apples?

Solution:

Revenue = $100 \times 2 = \$200$.

Answer: \$200

82. Question: What is the total revenue from Bananas?

Solution:

$$\text{Revenue} = 150 \times 1 = \$150.$$

Answer: \$150

83. Question: What is the total revenue from Oranges?

Solution:

$$\text{Revenue} = 200 \times 1.5 = \$300.$$

Answer: \$300

84. Question: What is the total revenue from all fruits?

Solution:

$$\text{Total} = 200 + 150 + 300 = \$650.$$

Answer: \$650

85. Question: Which fruit generated the highest revenue?

Solution:

Apples (\$200), Bananas (\$150), Oranges (\$300). Highest is Oranges.

Answer: Oranges

86. Question: What is the average price per kg of the fruits sold?

Solution:

Total revenue = \$650. Total kg = 100 + 150 + 200 = 450. Average price = $650 \div 450 = \$1.44/\text{kg}$.

Answer: \$1.44/kg

87. Question: What is the ratio of Apples sold to Bananas sold (in kg)?

Solution:

$$\text{Apples (100) : Bananas (150)} = 100 : 150 = 2 : 3.$$

Answer: 2 : 3

88. Question: How much more revenue did Oranges generate than Bananas?

Solution:

$$\text{Oranges (\$300) - Bananas (\$150)} = \$150.$$

Answer: \$150

89. Question: What percentage of total revenue came from Apples?

Solution:

$$\text{Apples (\$200)} \div \text{Total (\$650)} \times 100 = 30.77\%.$$

Answer: 30.77%

90. Question: What is the total weight of fruits sold?

Solution:

Total = $100 + 150 + 200 = 450$ kg.

Answer: 450 kg

Caselet 2: A School's Sports Day Points

- Team A: 50 points (Athletics), 30 points (Swimming), 20 points (Others)
- Team B: 40 points (Athletics), 20 points (Swimming), 30 points (Others)
- Team C: 30 points (Athletics), 40 points (Swimming), 10 points (Others)

91. Question: What is Team A's total points?

Solution:

Total = $50 + 30 + 20 = 100$ points.

Answer: 100 points

92. Question: Which team scored the most points in Swimming?

Solution:

Swimming: A (30), B (20), C (40). Highest is C (40).

Answer: Team C

93. Question: What is the average points scored by Team B?

Solution:

Total = $40 + 20 + 30 = 90$. Average = $90 \div 3 = 30$ points.

Answer: 30 points

94. Question: By how many points did Team A outscore Team C in Athletics?

Solution:

A (50) - C (30) = 20 points.

Answer: 20 points

95. Question: What is the total points scored in Others across all teams?

Solution:

Others: A (20) + B (30) + C (10) = 60 points.

Answer: 60 points

96. Question: Which team scored the fewest points overall?

Solution:

A: 100, B: 90, C: $30 + 40 + 10 = 80$. Lowest is C (80).

Answer: Team C

97. Question: What is the percentage of Team A's points from Athletics?

Solution:

Athletics (50) $\div$ Total (100) $\times 100 = 50\%$.

Answer: 50%

98. Question: What is the ratio of Team B's Swimming to Others points?

Solution:

Swimming (20) : Others (30) = 20 : 30 = 2 : 3.

Answer: 2 : 3

99. Question: What is the total points scored by Teams A and B in Swimming?

Solution:

A (30) + B (20) = 50 points.

Answer: 50 points

100. Question: Which team won the sports day (highest total points)?

Solution:

A: 100, B: 90, C: 80. Highest is A.

Answer: Team A

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SECTION: 02 Intermediate

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100 Data Interpretation Practice Questions (Intermediate Level) with Solutions

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Table-Based Questions (1?20)

Table 1: Quarterly Revenue (in \$1000) of Four Stores (Q1?Q4, 2023)

Store	Q1	Q2	Q3	Q4
-----	-----	-----	-----	-----
W	200	250	300	350
X	180	210	240	270
Y	150	160	170	180
Z	220	230	240	250

1. Question: What is the total annual revenue of Store W in 2023?

Solution:

Total = 200 + 250 + 300 + 350 = 1100 (\$1000) = \$1,100,000.

Answer: \$1,100,000

2. Question: Which store had the highest revenue in Q3?

Solution:

Q3: W (300), X (240), Y (170), Z (240). Highest is W (300).

Answer: Store W

3. Question: What is the average quarterly revenue of Store X?

Solution:

Total = $180 + 210 + 240 + 270 = 900$. Average = $900 \div 4 = 225$ (\$1000) = \$225,000.

Answer: \$225,000

4. Question: What is the percentage growth in Store Y's revenue from Q1 to Q4?

Solution:

Increase = $180 - 150 = 30$. % Growth = $(30 \div 150) \times 100 = 20\%$.

Answer: 20%

5. Question: What is the total revenue of all stores in Q2?

Solution:

Q2: $W (250) + X (210) + Y (160) + Z (230) = 850$ (\$1000) = \$850,000.

Answer: \$850,000

6. Question: What is the ratio of Store Z's revenue in Q1 to Q4?

Solution:

Q1 (220) : Q4 (250) = $220 : 250 = 22 : 25$.

Answer: 22 : 25

7. Question: By how much did Store W's revenue exceed Store X's in Q4?

Solution:

$W (350) - X (270) = 80$ (\$1000) = \$80,000.

Answer: \$80,000

8. Question: Which store showed the highest percentage growth from Q1 to Q2?

Solution:

W: $(250 - 200) \div 200 \times 100 = 25\%$.

X: $(210 - 180) \div 180 \times 100 = 16.67\%$.

Y: $(160 - 150) \div 150 \times 100 = 6.67\%$.

Z: $(230 - 220) \div 220 \times 100 = 4.55\%$.

Highest is W.

Answer: Store W

9. Question: What is the combined revenue of Stores W and Y in Q3?

Solution:

$W (300) + Y (170) = 470$ (\$1000) = \$470,000.

Answer: \$470,000

10. Question: Which quarter had the lowest total revenue across all stores?

Solution:

$$Q1: 200 + 180 + 150 + 220 = 750.$$

$$Q2: 250 + 210 + 160 + 230 = 850.$$

$$Q3: 300 + 240 + 170 + 240 = 950.$$

$$Q4: 350 + 270 + 180 + 250 = 1050 (\$1000).$$

Lowest is Q1 (\$750,000).

Answer: Q1

Table 2: Employees in Three Departments Across Two Branches

Department	Branch A (2022)	Branch A (2023)	Branch B (2022)	Branch B (2023)
HR	20	25	15	18
IT	50	60	40	45
Sales	30	35	25	30

11. Question: What is the total number of employees in Branch A in 2023?

Solution:

$$\text{Total} = 25 + 60 + 35 = 120 \text{ employees.}$$

Answer: 120 employees

12. Question: Which department in Branch B had the highest growth from 2022 to 2023?

Solution:

$$\text{HR: } (18 - 15) \div 15 \times 100 = 20\%.$$

$$\text{IT: } (45 - 40) \div 40 \times 100 = 12.5\%.$$

$$\text{Sales: } (30 - 25) \div 25 \times 100 = 20\%.$$

HR and Sales tie for highest.

Answer: HR or Sales

13. Question: What is the average number of IT employees across both branches in 2022?

Solution:

$$\text{Total} = 50 (\text{A}) + 40 (\text{B}) = 90. \text{ Average} = 90 \div 2 = 45 \text{ employees.}$$

Answer: 45 employees

14. Question: What is the percentage increase in Sales employees in Branch A from 2022 to 2023?

Solution:

$$\text{Increase} = 35 - 30 = 5. \text{ \% Increase} = (5 \div 30) \times 100 = 16.67\%.$$

Answer: 16.67%

15. Question: What is the total number of employees in HR across both branches in 2023?

Solution:

$$\text{HR: } 25 (\text{A}) + 18 (\text{B}) = 43 \text{ employees.}$$

Answer: 43 employees

16. Question: What is the ratio of IT employees in Branch A to Branch B in 2023?

Solution:

$$A (60) : B (45) = 60 : 45 = 4 : 3.$$

Answer: 4 : 3

17. Question: By how many employees did Branch A's total workforce grow from 2022 to 2023?

Solution:

$$2022: 20 + 50 + 30 = 100. 2023: 120. \text{Growth} = 120 - 100 = 20 \text{ employees.}$$

Answer: 20 employees

18. Question: Which branch had more employees in 2022?

Solution:

$$A: 100. B: 15 + 40 + 25 = 80. A \text{ is higher.}$$

Answer: Branch A

19. Question: What is the combined number of Sales employees in both branches in 2022?

Solution:

$$A (30) + B (25) = 55 \text{ employees.}$$

Answer: 55 employees

20. Question: What is the percentage of HR employees in Branch A (2023) relative to its total workforce?

Solution:

$$HR (25) \div Total (120) \times 100 = 20.83\%.$$

Answer: 20.83%

Bar Graph-Based Questions (21-40)

Bar Graph 1: Annual Production (in tons) of Three Crops (2020-2022)

- 2020: Rice (500), Wheat (400), Corn (300)

- 2021: Rice (550), Wheat (420), Corn (320)

- 2022: Rice (600), Wheat (450), Corn (350)

21. Question: What is the total production of Rice over the three years?

Solution:

$$Total = 500 + 550 + 600 = 1650 \text{ tons.}$$

Answer: 1650 tons

22. Question: Which crop had the highest production in 2021?

Solution:

$$2021: \text{Rice (550), Wheat (420), Corn (320). Highest is Rice.}$$

Answer: Rice

23. Question: What is the average annual production of Wheat?

Solution:

Total = $400 + 420 + 450 = 1270$. Average = $1270 \div 3 = 423.33$ tons.

Answer: 423.33 tons

24. Question: What is the percentage growth in Corn production from 2020 to 2022?

Solution:

Increase = $350 - 300 = 50$. % Growth = $(50 \div 300) \times 100 = 16.67\%$.

Answer: 16.67%

25. Question: What is the total production of all crops in 2020?

Solution:

2020: $500 + 400 + 300 = 1200$ tons.

Answer: 1200 tons

26. Question: What is the ratio of Rice to Wheat production in 2022?

Solution:

Rice (600) : Wheat (450) = $600 : 450 = 4 : 3$.

Answer: 4 : 3

27. Question: By how much did Rice production exceed Corn in 2021?

Solution:

Rice (550) - Corn (320) = 230 tons.

Answer: 230 tons

28. Question: Which crop showed the highest percentage growth from 2020 to 2021?

Solution:

Rice: $(550 - 500) \div 500 \times 100 = 10\%$.

Wheat: $(420 - 400) \div 400 \times 100 = 5\%$.

Corn: $(320 - 300) \div 300 \times 100 = 6.67\%$.

Highest is Rice.

Answer: Rice

29. Question: What is the combined production of Wheat and Corn in 2022?

Solution:

Wheat (450) + Corn (350) = 800 tons.

Answer: 800 tons

30. Question: Which year had the highest total production?

Solution:

2020: 1200. 2021: $550 + 420 + 320 = 1290$. 2022: $600 + 450 + 350 = 1400$ tons. Highest is 2022.

Answer: 2022

Bar Graph 2: Monthly Website Traffic (in thousands) for Two Sites (Jan-Mar)

- Jan: Site A (100), Site B (80)

- Feb: Site A (120), Site B (90)

- Mar: Site A (150), Site B (100)

31. Question: What is the total traffic for Site A over the three months?

Solution:

Total = $100 + 120 + 150 = 370$ (thousand) = 370,000 visitors.

Answer: 370,000 visitors

32. Question: Which site had higher traffic in March?

Solution:

Mar: A (150), B (100). A is higher.

Answer: Site A

33. Question: What is the average monthly traffic for Site B?

Solution:

Total = $80 + 90 + 100 = 270$. Average = $270 \div 3 = 90$ (thousand) = 90,000 visitors.

Answer: 90,000 visitors

34. Question: What is the percentage growth in Site A's traffic from Jan to Feb?

Solution:

Increase = $120 - 100 = 20$. % Growth = $(20 \div 100) \times 100 = 20\%$.

Answer: 20%

35. Question: What is the total traffic for both sites in Feb?

Solution:

Feb: A (120) + B (90) = 210 (thousand) = 210,000 visitors.

Answer: 210,000 visitors

36. Question: What is the ratio of Site A to Site B traffic in Mar?

Solution:

A (150) : B (100) = $150 : 100 = 3 : 2$.

Answer: 3 : 2

37. Question: By how much did Site A's traffic exceed Site B's in Jan?

Solution:

A (100) - B (80) = 20 (thousand) = 20,000 visitors.

Answer: 20,000 visitors

38. Question: Which site showed higher percentage growth from Jan to Mar?

Solution:

A: $(150 - 100) \div 100 \times 100 = 50\%$.

B: $(100 - 80) \div 80 \times 100 = 25\%$.

A is higher.

Answer: Site A

39. Question: What is the combined traffic of both sites in Jan?

Solution:

Jan: $A (100) + B (80) = 180$ (thousand) = 180,000 visitors.

Answer: 180,000 visitors

40. Question: Which month had the highest total traffic?

Solution:

Jan: 180. Feb: 210. Mar: $150 + 100 = 250$ (thousand). Highest is Mar.

Answer: March

Pie Chart-Based Questions (41-60)

Pie Chart 1: Market Share of Five Smartphone Brands (Total: 1000 units)

- Brand A: 30%
- Brand B: 25%
- Brand C: 20%
- Brand D: 15%
- Brand E: 10%

41. Question: How many units did Brand A sell?

Solution:

$A = 30\% \text{ of } 1000 = (30 \div 100) \times 1000 = 300$ units.

Answer: 300 units

42. Question: What is the combined market share of Brands B and C?

Solution:

$B (25\%) + C (20\%) = 45\%$.

Answer: 45%

43. Question: How many more units did Brand A sell than Brand D?

Solution:

$A (30\% = 300) - D (15\% = 150) = 300 - 150 = 150$ units.

Answer: 150 units

44. Question: What is the ratio of Brand C to Brand E units sold?

Solution:

$C (20\% = 200) : E (10\% = 100) = 200 : 100 = 2 : 1$.

Answer: 2 : 1

45. Question: What is the total number of units sold by Brands A, B, and C?

Solution:

$A (300) + B (25\% = 250) + C (200) = 750$ units.

Answer: 750 units

46. Question: Which brand has the smallest market share?

Solution:

E (10%) is smallest.

Answer: Brand E

47. Question: What is the percentage of Brand B's units relative to total units?

Solution:

$B = 25\%$ (given).

Answer: 25%

48. Question: How many units did Brand D sell?

Solution:

$D = 15\% \text{ of } 1000 = 150 \text{ units.}$

Answer: 150 units

49. Question: What is the difference in units sold between Brands A and E?

Solution:

$A (300) - E (10\% = 100) = 200 \text{ units.}$

Answer: 200 units

50. Question: If Brand A's sales increased by 10%, how many units would it sell?

Solution:

$A = 300. \text{ Increase} = 10\% \text{ of } 300 = 30. \text{ New sales} = 300 + 30 = 330 \text{ units.}$

Answer: 330 units

Pie Chart 2: Expenditure of a Family (Total: \$5000)

- Food: 35%

- Rent: 30%

- Education: 20%

- Transport: 10%

- Savings: 5%

51. Question: How much is spent on Food?

Solution:

$\text{Food} = 35\% \text{ of } 5000 = (35 \div 100) \times 5000 = \$1750.$

Answer: \$1750

52. Question: What is the combined expenditure on Rent and Education?

Solution:

$\text{Rent (30\%)} + \text{Education (20\%)} = 50\%. 50\% \text{ of } 5000 = \$2500.$

Answer: \$2500

53. Question: How much more is spent on Food than on Transport?

Solution:

$\text{Food (1750)} - \text{Transport (10\% = 500)} = 1750 - 500 = \$1250.$

Answer: \$1250

54. Question: What is the ratio of Education to Savings expenditure?

Solution:

Education (20% = 1000) : Savings (5% = 250) = 1000 : 250 = 4 : 1.

Answer: 4 : 1

55. Question: What is the total expenditure on Food, Rent, and Transport?

Solution:

Food (1750) + Rent (30% = 1500) + Transport (500) = 1750 + 1500 + 500 = \$3750.

Answer: \$3750

56. Question: Which category has the least expenditure?

Solution:

Savings (5%) is smallest.

Answer: Savings

57. Question: What percentage of expenditure is on Rent?

Solution:

Rent = 30% (given).

Answer: 30%

58. Question: How much is spent on Savings?

Solution:

Savings = 5% of 5000 = \$250.

Answer: \$250

59. Question: What is the difference between Food and Education expenditure?

Solution:

Food (1750) - Education (20% = 1000) = 1750 - 1000 = \$750.

Answer: \$750

60. Question: If Transport expenditure increased by 20%, what would be the new amount?

Solution:

Transport = 500. Increase = 20% of 500 = 100. New amount = 500 + 100 = \$600.

Answer: \$600

Line Graph-Based Questions (61-80)

Line Graph 1: Monthly Sales (in units) of Two Products (Jan-Jun)

- Jan: P1 (200), P2 (150)

- Feb: P1 (220), P2 (160)

- Mar: P1 (250), P2 (180)

- Apr: P1 (270), P2 (200)

- May: P1 (300), P2 (220)

- Jun: P1 (320), P2 (240)

61. Question: What is the total sales of P1 over the six months?

Solution:

$$\text{Total} = 200 + 220 + 250 + 270 + 300 + 320 = 1560 \text{ units.}$$

Answer: 1560 units

62. Question: Which product had higher sales in April?

Solution:

Apr: P1 (270), P2 (200). P1 is higher.

Answer: P1

63. Question: What is the average monthly sales of P2?

Solution:

$$\text{Total} = 150 + 160 + 180 + 200 + 220 + 240 = 1150. \text{ Average} = 1150 \div 6 = 191.67 \text{ units.}$$

Answer: 191.67 units

64. Question: What is the percentage growth in P1 sales from Feb to Mar?

Solution:

$$\text{Increase} = 250 - 220 = 30. \text{ \% Growth} = (30 \div 220) \times 100 = 13.64\%.$$

Answer: 13.64%

65. Question: What is the total sales of both products in May?

Solution:

$$\text{May: P1 (300) + P2 (220) = 520 units.}$$

Answer: 520 units

66. Question: What is the ratio of P1 to P2 sales in Jun?

Solution:

$$\text{P1 (320) : P2 (240) = } 320 : 240 = 4 : 3.$$

Answer: 4 : 3

67. Question: By how many units did P1 sales exceed P2 in Mar?

Solution:

$$\text{P1 (250) - P2 (180) = 70 units.}$$

Answer: 70 units

68. Question: Which product showed higher percentage growth from Jan to Jun?

Solution:

$$\text{P1: } (320 - 200) \div 200 \times 100 = 60\%.$$

$$\text{P2: } (240 - 150) \div 150 \times 100 = 60\%.$$

Both are equal.

Answer: Both P1 and P2

69. Question: What is the combined sales of both products in Jan?

Solution:

$$\text{Jan: P1 (200) + P2 (150) = 350 units.}$$

Answer: 350 units

70. Question: Which month had the highest total sales?

Solution:

Jan: 350. Feb: $220 + 160 = 380$. Mar: $250 + 180 = 430$. Apr: $270 + 200 = 470$.

May: 520. Jun: $320 + 240 = 560$ units. Highest is Jun.

Answer: June

Line Graph 2: Quarterly Profit (\$1000) of a Company (Q1?Q4, 2022?2023)

- 2022: Q1 (50), Q2 (60), Q3 (70), Q4 (80)

- 2023: Q1 (55), Q2 (65), Q3 (75), Q4 (90)

71. Question: What is the total profit in 2023?

Solution:

Total = $55 + 65 + 75 + 90 = 285$ (\$1000) = \$285,000.

Answer: \$285,000

72. Question: Which quarter had the highest profit in 2023?

Solution:

Q4 (90) is highest.

Answer: Q4

73. Question: What is the average quarterly profit in 2022?

Solution:

Total = $50 + 60 + 70 + 80 = 260$. Average = $260 \div 4 = 65$ (\$1000) = \$65,000.

Answer: \$65,000

74. Question: What is the percentage growth in profit from Q1 to Q2 in 2023?

Solution:

Increase = $65 - 55 = 10$. % Growth = $(10 \div 55) \times 100 \approx 18.18\%$.

Answer: 18.18%

75. Question: What is the total profit in Q3 across both years?

Solution:

2022 (70) + 2023 (75) = 145 (\$1000) = \$145,000.

Answer: \$145,000

76. Question: What is the ratio of profit in Q2 (2022) to Q2 (2023)?

Solution:

2022 (60) : 2023 (65) = $60 : 65 = 12 : 13$.

Answer: 12 : 13

77. Question: By how much did profit increase from Q4 2022 to Q4 2023?

Solution:

2023 (90) - 2022 (80) = 10 (\$1000) = \$10,000.

Answer: \$10,000

78. Question: Which year had higher total profit?

Solution:

2022: 260. 2023: 285 (\$1000). 2023 is higher.

Answer: 2023

79. Question: What is the combined profit in Q1 and Q2 of 2023?

Solution:

Q1 (55) + Q2 (65) = 120 (\$1000) = \$120,000.

Answer: \$120,000

80. Question: Which quarter showed the highest percentage growth from 2022 to 2023?

Solution:

Q1: $(55 - 50) \div 50 \times 100 = 10\%$.

Q2: $(65 - 60) \div 60 \times 100 = 8.33\%$.

Q3: $(75 - 70) \div 70 \times 100 = 7.14\%$.

Q4: $(90 - 80) \div 80 \times 100 = 12.5\%$.

Highest is Q4.

Answer: Q4

Caselet-Based Questions (81-100)

Caselet 1: A Company Sells Three Products

- Product X: 500 units at \$10/unit, 20% discount
- Product Y: 400 units at \$15/unit, 10% discount
- Product Z: 300 units at \$20/unit, no discount

81. Question: What is the revenue from Product X after discount?

Solution:

Discount = 20% of \$10 = \$2. Price = \$8/unit. Revenue = $500 \times 8 = \$4000$.

Answer: \$4000

82. Question: What is the revenue from Product Y after discount?

Solution:

Discount = 10% of \$15 = \$1.5. Price = \$13.5/unit. Revenue = $400 \times 13.5 = \$5400$.

Answer: \$5400

83. Question: What is the revenue from Product Z?

Solution:

Revenue = $300 \times 20 = \$6000$.

Answer: \$6000

84. Question: What is the total revenue from all products?

Solution:

Total = $4000 + 5400 + 6000 = \$15,400$.

Answer: \$15,400

85. Question: Which product generated the highest revenue?

Solution:

X (\$4000), Y (\$5400), Z (\$6000). Highest is Z.

Answer: Product Z

86. Question: What is the average price per unit sold after discounts?

Solution:

Total units = $500 + 400 + 300 = 1200$. Total revenue = \$15,400. Average = $15,400 \div 1200 = \$12.83/\text{unit}$.

Answer: \$12.83/unit

87. Question: What is the ratio of units sold of X to Y?

Solution:

X (500) : Y (400) = $500 : 400 = 5 : 4$.

Answer: 5 : 4

88. Question: How much more revenue did Z generate than X?

Solution:

Z (\$6000) - X (\$4000) = \$2000.

Answer: \$2000

89. Question: What percentage of total revenue comes from Y?

Solution:

Y (\$5400) $\div$ Total (\$15,400) $\times 100 = 35.06\%$.

Answer: 35.06%

90. Question: What is the total discount amount given on X and Y?

Solution:

X: $500 \times \$2 = \1000 . Y: $400 \times \$1.5 = \600 . Total = $1000 + 600 = \$1600$.

Answer: \$1600

Caselet 2: A School's Exam Results

- Class A: 50 students, 60% passed, average score 70
- Class B: 40 students, 75% passed, average score 65
- Class C: 30 students, 80% passed, average score 80

91. Question: How many students passed in Class A?

Solution:

Passed = 60% of 50 = $(60 \div 100) \times 50 = 30$ students.

Answer: 30 students

92. Question: What is the total score of all students in Class B?

Solution:

$$\text{Total score} = 40 \times 65 = 2600.$$

Answer: 2600

93. Question: Which class had the highest pass percentage?

Solution:

A: 60%, B: 75%, C: 80%. Highest is C.

Answer: Class C

94. Question: How many students failed in Class C?

Solution:

$$\text{Passed} = 80\% \text{ of } 30 = 24. \text{ Failed} = 30 - 24 = 6 \text{ students.}$$

Answer: 6 students

95. Question: What is the total number of students who passed across all classes?

Solution:

$$\text{A: } 30, \text{ B: } 75\% \text{ of } 40 = 30, \text{ C: } 24. \text{ Total} = 30 + 30 + 24 = 84 \text{ students.}$$

Answer: 84 students

96. Question: What is the average score of all students across the three classes?

Solution:

$$\text{Total students} = 50 + 40 + 30 = 120. \text{ Total score} = (50 \times 70) + (40 \times 65) + (30 \times 80) = 3500 + 2600 + 2400 = 8500.$$

$$\text{Average} = 8500 \div 120 = 70.83.$$

Answer: 70.83

97. Question: What is the ratio of passed students in Class A to Class B?

Solution:

$$\text{A (30) : B (30)} = 30 : 30 = 1 : 1.$$

Answer: 1 : 1

98. Question: How much higher is Class C's average score than Class B's?

Solution:

$$\text{C (80) - B (65)} = 15.$$

Answer: 15

99. Question: What percentage of total students passed?

Solution:

$$\text{Total students} = 120. \text{ Passed} = 84. \% \text{ Passed} = (84 \div 120) \times 100 = 70\%.$$

Answer: 70%

100. Question: What is the total score of passed students in Class C?

Solution:

$$\text{Passed} = 24 \text{ students. Total score} = 24 \times 80 = 1920.$$

Answer: 1920

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SECTION: 03 Advance

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--- README.md ---

100 Data Interpretation Practice Questions (Advanced Level) with Solutions

This file contains 100 advanced-level practice questions for Data Interpretation, designed to rigorously test your ability to analyze complex datasets for fresher job aptitude tests, especially for AI/ML roles. Each question includes a dataset (multiple tables, stacked bar graphs, combination charts, pie charts with secondary data, or caselets) and a detailed solution with step-by-step calculations. Topics cover compound growth rates, weighted averages, trend analysis, multi-dataset integration, percentages, ratios, and advanced comparisons.

Multiple Table-Based Questions (1?20)

Table 1: Annual Sales (in \$1000) of Three Products Across Two Regions (2021?2023)

| Product | Region A (2021) | Region A (2022) | Region A (2023) | Region B (2021) | Region B (2022) | Region B (2023) |

	Region A (2021)	Region A (2022)	Region A (2023)	Region B (2021)	Region B (2022)	Region B (2023)
X	500	550	600	400	450	500
Y	300	320	350	250	270	300
Z	200	220	250	150	180	200

Table 2: Unit Price (\$ per unit) of Products (2021?2023)

| Product | 2021 | 2022 | 2023 |

	2021	2022	2023
X	10	12	15
Y	8	9	10
Z	5	6	7

1. Question: What is the total revenue from Product X in Region A in 2023?

Solution:

Sales = 600 (\$1000) = 600,000. Units = $600,000 \div 15 = 40,000$. Revenue = $40,000 \times 15 = \$600,000$.

Answer: \$600,000

2. Question: Which product had the highest revenue in Region B in 2022?

Solution:

X: $450,000 \div 12 = 37,500$ units $\times 12 = \$450,000$.

Y: $270,000 \div 9 = 30,000$ units $\times 9 = \$270,000$.

Z: $180,000 \div 6 = 30,000 \text{ units} \times 6 = \$180,000$.

Highest is X.

Answer: Product X

3. Question: What is the compound annual growth rate (CAGR) of Product Y's sales in Region A from 2021 to 2023?

Solution:

Initial (2021) = 300 (\$1000), Final (2023) = 350 (\$1000).

$\text{CAGR} = [(350 \div 300)^{(1/2)} - 1] \times 100 \approx [(1.1667)^{0.5} - 1] \times 100 \approx 8.01\%$.

Answer: 8.01%

4. Question: What is the weighted average unit price of all products in 2022, weighted by units sold in Region A?

Solution:

Units X: $550,000 \div 12 \approx 45,833$. Y: $320,000 \div 9 \approx 35,556$. Z: $220,000 \div 6 \approx 36,667$.

Total units = $45,833 + 35,556 + 36,667 \approx 118,056$.

Weighted price = $[(45,833 \times 12) + (35,556 \times 9) + (36,667 \times 6)] \div 118,056 \approx \9.26 .

Answer: \$9.26

5. Question: What is the total revenue across both regions for Product Z in 2021?

Solution:

A: $200,000 \div 5 = 40,000 \text{ units} \times 5 = \$200,000$.

B: $150,000 \div 5 = 30,000 \text{ units} \times 5 = \$150,000$.

Total = $200,000 + 150,000 = \$350,000$.

Answer: \$350,000

6. Question: What is the percentage contribution of Product X to total revenue in Region B in 2023?

Solution:

X: $500,000 \div 15 \approx 33,333 \text{ units} \times 15 = \$500,000$.

Y: $300,000 \div 10 = 30,000 \text{ units} \times 10 = \$300,000$.

Z: $200,000 \div 7 \approx 28,571 \text{ units} \times 7 \approx \$200,000$.

Total = $500,000 + 300,000 + 200,000 = \$1,000,000$.

% X = $(500,000 \div 1,000,000) \times 100 = 50\%$.

Answer: 50%

7. Question: What is the ratio of units sold of Product Y in Region A to Region B in 2022?

Solution:

A: $320,000 \div 9 \approx 35,556 \text{ units}$. B: $270,000 \div 9 = 30,000 \text{ units}$.

Ratio = $35,556 : 30,000 \approx 1.19 : 1 \approx 119 : 100$.

Answer: 119 : 100

8. Question: Which region showed the highest revenue growth for Product Z from 2021 to 2023?

Solution:

A: 2021: 200,000, 2023: 250,000. % Growth = $(50 \div 200) \times 100 = 25\%$.

B: 2021: 150,000, 2023: 200,000. % Growth = $(50 \div 150) \times 100 \approx 33.33\%$.

B is higher.

Answer: Region B

9. Question: What is the total revenue of all products in Region A in 2022?

Solution:

X: 550,000. Y: 320,000. Z: 220,000. Total = $550,000 + 320,000 + 220,000 = \$1,090,000$.

Answer: \$1,090,000

10. Question: What is the percentage increase in unit price of Product X from 2021 to 2023?

Solution:

Increase = $15 - 10 = 5$. % Increase = $(5 \div 10) \times 100 = 50\%$.

Answer: 50%

Table 3: Production Costs (\$ per unit) and Units Produced (2021-2023)

Product	Cost (2021)	Cost (2022)	Cost (2023)	Units (2021)	Units (2022)	Units (2023)
P	4	5	6	20,000	22,000	25,000
Q	6	7	8	15,000	18,000	20,000

11. Question: What is the total production cost of Product P in 2022?

Solution:

Cost = $22,000 \times 5 = \$110,000$.

Answer: \$110,000

12. Question: What is the percentage increase in production cost per unit for Product Q from 2021 to 2023?

Solution:

Increase = $8 - 6 = 2$. % Increase = $(2 \div 6) \times 100 \approx 33.33\%$.

Answer: 33.33%

13. Question: What is the total cost of producing both products in 2023?

Solution:

P: $25,000 \times 6 = \$150,000$. Q: $20,000 \times 8 = \$160,000$. Total = $150,000 + 160,000 = \$310,000$.

Answer: \$310,000

14. Question: What is the CAGR of units produced for Product P from 2021 to 2023?

Solution:

Initial = 20,000, Final = 25,000. CAGR = $[(25,000 \div 20,000)^{(1/2)} - 1] \times 100 \approx 11.80\%$.

Answer: 11.80%

15. Question: What is the ratio of total production costs of P to Q in 2021?

Solution:

P: $20,000 \times 4 = \$80,000$. Q: $15,000 \times 6 = \$90,000$. Ratio = $80,000 : 90,000 = 8 : 9$.

Answer: 8 : 9

16. Question: Which product had higher cost growth per unit from 2021 to 2022?

Solution:

P: $(5 - 4) \div 4 \times 100 = 25\%$. Q: $(7 - 6) \div 6 \times 100 = 16.67\%$. P is higher.

Answer: Product P

17. Question: What is the total units produced in 2022?

Solution:

P: 22,000 + Q: 18,000 = 40,000 units.

Answer: 40,000 units

18. Question: What is the percentage contribution of Product Q's cost to total cost in 2023?

Solution:

Q: \$160,000. Total: \$310,000. $\% = (160,000 \div 310,000) \times 100 = 51.61\%$.

Answer: 51.61%

19. Question: What is the difference in production cost for Product P between 2021 and 2023?

Solution:

2021: $20,000 \times 4 = \$80,000$. 2023: $25,000 \times 6 = \$150,000$. Difference = $150,000 - 80,000 = \$70,000$.

Answer: \$70,000

20. Question: What is the weighted average cost per unit in 2022, weighted by units produced?

Solution:

Total units = 40,000. Cost = $(22,000 \times 5) + (18,000 \times 7) = 110,000 + 126,000 = \$236,000$.

Average = $236,000 \div 40,000 = \$5.90$.

Answer: \$5.90

Stacked Bar Graph-Based Questions (21-40)

Stacked Bar Graph 1: Revenue (\$1000) by Product Category in Three Cities (2021-2023)

- 2021: City A (Electronics: 300, Apparel: 200, Food: 100), City B (200, 150, 50), City C (250, 100, 50)

- 2022: City A (350, 220, 120), City B (220, 160, 60), City C (270, 120, 60)

- 2023: City A (400, 250, 150), City B (250, 180, 70), City C (300, 150, 80)

21. Question: What is the total revenue from Electronics in City A over the three years?

Solution:

Total = $300 + 350 + 400 = 1050$ (\$1000) = \$1,050,000.

Answer: \$1,050,000

22. Question: Which city had the highest Apparel revenue in 2023?

Solution:

A: 250, B: 180, C: 150. Highest is A.

Answer: City A

23. Question: What is the percentage growth of Food revenue in City B from 2021 to 2023?

Solution:

$$\text{Increase} = 70 - 50 = 20. \text{ \% Growth} = (20 \div 50) \times 100 = 40\%.$$

Answer: 40%

24. Question: What is the total revenue in City C in 2022?

Solution:

$$\text{Total} = 270 + 120 + 60 = 450 (\$1000) = \$450,000.$$

Answer: \$450,000

25. Question: What is the ratio of Electronics to Apparel revenue in City A in 2021?

Solution:

$$\text{Electronics (300) : Apparel (200)} = 300 : 200 = 3 : 2.$$

Answer: 3 : 2

26. Question: Which category showed the highest growth in City C from 2021 to 2023?

Solution:

$$\text{Electronics: } (300 - 250) \div 250 \times 100 = 20\%.$$

$$\text{Apparel: } (150 - 100) \div 100 \times 100 = 50\%.$$

$$\text{Food: } (80 - 50) \div 50 \times 100 = 60\%.$$

Highest is Food.

Answer: Food

27. Question: What is the percentage contribution of Electronics to total revenue in City B in 2023?

Solution:

$$\text{Total} = 250 + 180 + 70 = 500. \text{ \% Electronics} = (250 \div 500) \times 100 = 50\%.$$

Answer: 50%

28. Question: What is the combined revenue of Apparel and Food in City A in 2022?

Solution:

$$\text{Apparel (220) + Food (120)} = 340 (\$1000) = \$340,000.$$

Answer: \$340,000

29. Question: Which year had the highest total revenue across all cities?

Solution:

$$2021: A (600) + B (400) + C (400) = 1400.$$

$$2022: A (690) + B (440) + C (450) = 1580.$$

$$2023: A (800) + B (500) + C (530) = 1830 (\$1000).$$

Highest is 2023.

Answer: 2023

30. Question: What is the CAGR of total revenue in City A from 2021 to 2023?

Solution:

Initial = 600, Final = 800. $CAGR = [(800 \div 600)^{(1/2)} - 1] \times 100 \approx 15.47\%$.

Answer: 15.47%

Stacked Bar Graph 2: Student Enrollments by Program (2021-2023)

- 2021: UG (500), PG (300), Diploma (200)

- 2022: UG (550), PG (320), Diploma (220)

- 2023: UG (600), PG (350), Diploma (250)

31. Question: What is the total enrollment in PG over the three years?

Solution:

Total = 300 + 320 + 350 = 970 students.

Answer: 970 students

32. Question: Which program had the highest enrollment in 2023?

Solution:

UG (600), PG (350), Diploma (250). Highest is UG.

Answer: UG

33. Question: What is the percentage growth of Diploma enrollment from 2021 to 2023?

Solution:

Increase = 250 - 200 = 50. % Growth = $(50 \div 200) \times 100 = 25\%$.

Answer: 25%

34. Question: What is the total enrollment in 2022?

Solution:

Total = 550 + 320 + 220 = 1090 students.

Answer: 1090 students

35. Question: What is the ratio of UG to PG enrollment in 2021?

Solution:

UG (500) : PG (300) = 500 : 300 = 5 : 3.

Answer: 5 : 3

36. Question: Which program showed the highest percentage growth from 2021 to 2022?

Solution:

UG: $(550 - 500) \div 500 \times 100 = 10\%$.

PG: $(320 - 300) \div 300 \times 100 \approx 6.67\%$.

Diploma: $(220 - 200) \div 200 \times 100 = 10\%$.

UG and Diploma tie.

Answer: UG or Diploma

37. Question: What is the percentage contribution of PG to total enrollment in 2023?

Solution:

Total = 600 + 350 + 250 = 1200. % PG = $(350 \div 1200) \times 100 \approx 29.17\%$.

Answer: 29.17%

38. Question: What is the combined enrollment of UG and Diploma in 2022?

Solution:

UG (550) + Diploma (220) = 770 students.

Answer: 770 students

39. Question: Which year had the highest total enrollment?

Solution:

2021: 1000. 2022: 1090. 2023: 1200. Highest is 2023.

Answer: 2023

40. Question: What is the CAGR of UG enrollment from 2021 to 2023?

Solution:

Initial = 500, Final = 600. $CAGR = [(600 \div 500)^{(1/2)} - 1] \times 100 \approx 9.54\%$.

Answer: 9.54%

Combination Chart-Based Questions (41-60)

Combination Chart 1: Sales (units) and Profit Margin (%) of a Product (2019-2023)

- Bar (Sales): 2019 (10,000), 2020 (12,000), 2021 (15,000), 2022 (18,000), 2023 (20,000)

- Line (Profit Margin): 2019 (5%), 2020 (6%), 2021 (7%), 2022 (8%), 2023 (10%)

41. Question: What is the profit in 2023?

Solution:

Profit = Sales $\times$ Margin = 20,000 $\times$ (10 $\div$ 100) = 2000 units.

Answer: 2000 units

42. Question: Which year had the highest profit margin?

Solution:

2023 (10%) is highest.

Answer: 2023

43. Question: What is the CAGR of sales from 2019 to 2023?

Solution:

Initial = 10,000, Final = 20,000. $CAGR = [(20,000 \div 10,000)^{(1/4)} - 1] \times 100 \approx 18.92\%$.

Answer: 18.92%

44. Question: What is the total profit over the five years?

Solution:

2019: 10,000 $\times$ 0.05 = 500. 2020: 12,000 $\times$ 0.06 = 720. 2021: 15,000 $\times$ 0.07 = 1050.

2022: 18,000 $\times$ 0.08 = 1440. 2023: 2000. Total = 500 + 720 + 1050 + 1440 + 2000 = 5710 units.

Answer: 5710 units

45. Question: What is the ratio of profit in 2021 to 2022?

Solution:

2021: 1050. 2022: 1440. Ratio = 1050 : 1440 = 7 : 9.6 ? 35 : 48.

Answer: 35 : 48

46. Question: Which year showed the highest percentage growth in sales?

Solution:

2020: $(12,000 - 10,000) \div 10,000 \times 100 = 20\%$.

2021: $(15,000 - 12,000) \div 12,000 \times 100 = 25\%$.

2022: $(18,000 - 15,000) \div 15,000 \times 100 = 20\%$.

2023: $(20,000 - 18,000) \div 18,000 \times 100 = 11.11\%$.

Highest is 2021.

Answer: 2021

47. Question: What is the percentage contribution of 2023 profit to total profit?

Solution:

Total profit = 5710. 2023: 2000. % = $(2000 \div 5710) \times 100 = 35.03\%$.

Answer: 35.03%

48. Question: What is the average profit margin over the five years?

Solution:

Total = 5 + 6 + 7 + 8 + 10 = 36. Average = $36 \div 5 = 7.2\%$.

Answer: 7.2%

49. Question: What is the difference in profit between 2022 and 2023?

Solution:

2023: 2000 - 2022: 1440 = 560 units.

Answer: 560 units

50. Question: If sales in 2024 are projected to grow by 10% with a 12% margin, what is the projected profit?

Solution:

2023 Sales = 20,000. 2024 Sales = $20,000 \times 1.1 = 22,000$. Profit = $22,000 \times 0.12 = 2640$ units.

Answer: 2640 units

Combination Chart 2: Revenue (\$1000) and Growth Rate (%) of a Company (2020-2023)

- Bar (Revenue): 2020 (500), 2021 (600), 2022 (750), 2023 (900)

- Line (Growth Rate): 2021 (20%), 2022 (25%), 2023 (20%)

51. Question: What is the revenue in 2023?

Solution:

Revenue = 900 (\$1000) = \$900,000.

Answer: \$900,000

52. Question: Which year had the highest growth rate?

Solution:

2022 (25%) is highest.

Answer: 2022

53. Question: What is the CAGR of revenue from 2020 to 2023?

Solution:

Initial = 500, Final = 900. $CAGR = [(900 \div 500)^{(1/3)} - 1] \times 100 \approx 21.64\%$.

Answer: 21.64%

54. Question: What is the total revenue over the four years?

Solution:

Total = 500 + 600 + 750 + 900 = 2750 (\$1000) = \$2,750,000.

Answer: \$2,750,000

55. Question: What is the ratio of revenue in 2021 to 2022?

Solution:

2021 (600) : 2022 (750) = 600 : 750 = 4 : 5.

Answer: 4 : 5

56. Question: What is the absolute growth in revenue from 2020 to 2023?

Solution:

Growth = 900 - 500 = 400 (\$1000) = \$400,000.

Answer: \$400,000

57. Question: What is the percentage contribution of 2022 revenue to total revenue?

Solution:

2022: 750. Total: 2750. % = $(750 \div 2750) \times 100 \approx 27.27\%$.

Answer: 27.27%

58. Question: What is the average growth rate from 2021 to 2023?

Solution:

Total = 20 + 25 + 20 = 65. Average = $65 \div 3 \approx 21.67\%$.

Answer: 21.67%

59. Question: What is the difference in revenue between 2022 and 2023?

Solution:

2023: 900 - 2022: 750 = 150 (\$1000) = \$150,000.

Answer: \$150,000

60. Question: If 2024 revenue grows at 15%, what is the projected revenue?

Solution:

2023: 900. 2024 = $900 \times 1.15 = 1035$ (\$1000) = \$1,035,000.

Answer: \$1,035,000

Pie Chart with Secondary Data-Based Questions (61-80)

Pie Chart 1: Market Share (%) and Revenue (\$1000) of Four Companies in 2023

- Company A: 40%, Revenue: 800
- Company B: 30%, Revenue: 600
- Company C: 20%, Revenue: 400
- Company D: 10%, Revenue: 200

61. Question: What is the total market revenue in 2023?

Solution:

$$\text{Total} = 800 + 600 + 400 + 200 = 2000 (\$1000) = \$2,000,000.$$

Answer: \$2,000,000

62. Question: Which company has the highest market share?

Solution:

A (40%) is highest.

Answer: Company A

63. Question: What is the ratio of revenue of Company A to Company C?

Solution:

$$A (800) : C (400) = 800 : 400 = 2 : 1.$$

Answer: 2 : 1

64. Question: What is the percentage contribution of Company B's revenue to total revenue?

Solution:

$$B: 600. \text{ Total: } 2000. \% = (600 \div 2000) \times 100 = 30\%.$$

Answer: 30%

65. Question: If Company D's revenue grows by 25% in 2024, what is the new revenue?

Solution:

$$D: 200. \text{ Increase} = 200 \times 0.25 = 50. \text{ New} = 200 + 50 = 250 (\$1000) = \$250,000.$$

Answer: \$250,000

66. Question: What is the combined market share of Companies C and D?

Solution:

$$C (20\%) + D (10\%) = 30\%.$$

Answer: 30%

67. Question: What is the difference in revenue between Companies A and B?

Solution:

$$A (800) - B (600) = 200 (\$1000) = \$200,000.$$

Answer: \$200,000

68. Question: If total market revenue grows by 10% in 2024, what is Company A's projected revenue (same share)?

Solution:

$$\text{Total: } 2000 \times 1.1 = 2200. A: 40\% \text{ of } 2200 = 880 (\$1000) = \$880,000.$$

Answer: \$880,000

69. Question: What is the weighted average market share, weighted by revenue?

Solution:

Weighted share = $[(40 \times 800) + (30 \times 600) + (20 \times 400) + (10 \times 200)] \div 2000 = 40\%$.

Answer: 40%

70. Question: Which company has the lowest revenue?

Solution:

D (200) is lowest.

Answer: Company D

Pie Chart 2: Budget Allocation (%) and Actual Spending (\$1000) in 2023

- Salaries: 50%, Spending: 1000
- Rent: 20%, Spending: 400
- Supplies: 15%, Spending: 300
- Marketing: 15%, Spending: 300

71. Question: What is the total budget spending in 2023?

Solution:

Total = $1000 + 400 + 300 + 300 = 2000$ (\$1000) = \$2,000,000.

Answer: \$2,000,000

72. Question: Which category has the highest allocation?

Solution:

Salaries (50%) is highest.

Answer: Salaries

73. Question: What is the ratio of Salaries to Rent spending?

Solution:

Salaries (1000) : Rent (400) = $1000 : 400 = 5 : 2$.

Answer: 5 : 2

74. Question: What is the percentage contribution of Supplies to total spending?

Solution:

Supplies: 300. Total: 2000. % = $(300 \div 2000) \times 100 = 15\%$.

Answer: 15%

75. Question: If Marketing spending increases by 20% in 2024, what is the new amount?

Solution:

Marketing: 300. Increase = $300 \times 0.2 = 60$. New = $300 + 60 = 360$ (\$1000) = \$360,000.

Answer: \$360,000

76. Question: What is the combined allocation of Rent and Supplies?

Solution:

Rent (20%) + Supplies (15%) = 35%.

Answer: 35%

77. Question: What is the difference in spending between Salaries and Marketing?

Solution:

Salaries (1000) - Marketing (300) = 700 (\$1000) = \$700,000.

Answer: \$700,000

78. Question: If total spending grows by 5% in 2024, what is Salaries? projected spending (same allocation)?

Solution:

Total: $2000 \times 1.05 = 2100$. Salaries: 50% of 2100 = 1050 (\$1000) = \$1,050,000.

Answer: \$1,050,000

79. Question: What is the weighted average allocation, weighted by spending?

Solution:

Weighted = $[(50 \times 1000) + (20 \times 400) + (15 \times 300) + (15 \times 300)] \div 2000 = 50\%$.

Answer: 50%

80. Question: Which categories have equal spending?

Solution:

Supplies (300) and Marketing (300) are equal.

Answer: Supplies and Marketing

Caselet-Based Questions (81-100)

Caselet 1: A Retail Chain's Sales and Costs

- Store A: 1000 units sold at \$50/unit, 10% discount, cost \$30/unit

- Store B: 800 units sold at \$60/unit, 15% discount, cost \$35/unit

- Store C: 600 units sold at \$70/unit, no discount, cost \$40/unit

81. Question: What is the revenue from Store A after discount?

Solution:

Discount = 10% of 50 = 5. Price = $50 - 5 = 45$. Revenue = $1000 \times 45 = \$45,000$.

Answer: \$45,000

82. Question: What is the profit from Store B?

Solution:

Discount = 15% of 60 = 9. Price = $60 - 9 = 51$. Revenue = $800 \times 51 = \$40,800$.

Cost = $800 \times 35 = \$28,000$. Profit = $40,800 - 28,000 = \$12,800$.

Answer: \$12,800

83. Question: What is the total revenue from all stores?

Solution:

A: \$45,000. B: \$40,800. C: $600 \times 70 = \$42,000$. Total = $45,000 + 40,800 + 42,000 = \$127,800$.

Answer: \$127,800

84. Question: What is the total profit margin (%) across all stores?

Solution:

Total cost: A: $1000 \times 30 = 30,000$. B: 28,000. C: $600 \times 40 = 24,000$. Total = 82,000.

Profit = $127,800 - 82,000 = 45,800$. Margin = $(45,800 \div 127,800) \times 100 \approx 35.84\%$.

Answer: 35.84%

85. Question: Which store has the highest profit?

Solution:

A: $45,000 - 30,000 = 15,000$. B: \$12,800. C: $42,000 - 24,000 = 18,000$. Highest is C.

Answer: Store C

86. Question: What is the weighted average selling price per unit after discounts?

Solution:

Total units = $1000 + 800 + 600 = 2400$. Revenue = \$127,800. Average = $127,800 \div 2400 = 53.25$.

Answer: \$53.25

87. Question: What is the ratio of units sold in Store A to Store C?

Solution:

A (1000) : C (600) = $1000 : 600 = 5 : 3$.

Answer: 5 : 3

88. Question: What is the percentage contribution of Store B's profit to total profit?

Solution:

Total profit = 45,800. B: 12,800. % = $(12,800 \div 45,800) \times 100 \approx 27.95\%$.

Answer: 27.95%

89. Question: What is the total discount amount across all stores?

Solution:

A: $1000 \times 5 = 5000$. B: $800 \times 9 = 7200$. C: \$0. Total = $5000 + 7200 = 12,200$.

Answer: \$12,200

90. Question: If Store C's units sold increase by 20%, what is the new profit?

Solution:

New units = $600 \times 1.2 = 720$. Revenue = $720 \times 70 = 50,400$. Cost = $720 \times 40 = 28,800$.

Profit = $50,400 - 28,800 = 21,600$.

Answer: \$21,600

Caselet 2: A University's Research Funding and Publications

- Dept A: \$500,000 funding, 50 publications, 20% high-impact
- Dept B: \$400,000 funding, 40 publications, 25% high-impact
- Dept C: \$300,000 funding, 30 publications, 30% high-impact

91. Question: How many high-impact publications did Dept A produce?

Solution:

High-impact = 20% of 50 = 10 publications.

Answer: 10 publications

92. Question: What is the average funding per publication for Dept B?

Solution:

Funding = \$400,000. Publications = 40. Average = $400,000 \div 40 = \$10,000$.

Answer: \$10,000

93. Question: Which department had the highest number of high-impact publications?

Solution:

A: 10. B: 25% of 40 = 10. C: 30% of 30 = 9. A and B tie.

Answer: Dept A or B

94. Question: What is the total funding across all departments?

Solution:

Total = $500,000 + 400,000 + 300,000 = \$1,200,000$.

Answer: \$1,200,000

95. Question: What is the total number of publications across all departments?

Solution:

Total = $50 + 40 + 30 = 120$ publications.

Answer: 120 publications

96. Question: What is the weighted average funding per publication?

Solution:

Total funding = \$1,200,000. Total publications = 120. Average = $1,200,000 \div 120 = \$10,000$.

Answer: \$10,000

97. Question: What is the ratio of high-impact publications in Dept B to Dept C?

Solution:

B: 10. C: 9. Ratio = 10 : 9.

Answer: 10 : 9

98. Question: What is the percentage contribution of Dept C's funding to total funding?

Solution:

C: 300,000. Total: 1,200,000. % = $(300,000 \div 1,200,000) \times 100 = 25\%$.

Answer: 25%

99. Question: What is the total number of high-impact publications?

Solution:

A: 10, B: 10, C: 9. Total = $10 + 10 + 9 = 29$.

Answer: 29 publications

100. Question: If Dept A's funding increases by 10% and publications remain the same, what is the new funding per publication?

Solution:

New funding = $500,000 \times 1.1 = \$550,000$. Publications = 50. Average = $550,000 \div 50 = \$11,000$.

Answer: \$11,000